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15 August 1962
SUPERSEDING
MIL-STD-725 (NORD)
23 September 1959

MILITARY STANDARD

METHOD OF MARKING SCALES FOR SIGHTS AND FIRE CONTROL INSTRUMENTS

FSC 1220

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DEPARTMENT OF THE NAVY

BUREAU OF NAVAL WEAPONS

Washington 25, D.C.

Method of Marking Scales for Sights and Fire Control Instruments

MIL-STD-725A (WEPS)

1. This standard has been approved by the Bureau of Naval Weapons and is published to establish a Method of Marking Scales for Sights and Fire Control Instruments.

2. Use of this standard by activities under cognizance of the Bureau of Naval Weapons shall be mandatory effective on date of issue.

3. Recommended corrections, additions, or deletions should be addressed to the Chief of the Bureau of Naval Weapons, Department of the Navy, Washington 25, D.C.

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Department of the Navy
Naval Ordnance Station
Indian Head, Maryland 20640

Method of Marking Scales for Sights and Fire Control Instruments

MIL-STD-725A(OS)

1. This standard has been approved by the Naval Ordnance Systems Command and is published to establish a Method of Marking Scales for Sights and Fire Control Instruments.

2. Use of this standard by activities under cognizance of the Naval Ordnance Systems Command shall be mandatory, effective on the date of issue.

3. Recommended corrections, additions, or deletions shall be addressed to the Commanding Officer, Naval Ordnance Station, Indian Head, MD 20640, Attn: Standardization Division (QA3).

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CONTENTS

			Page
Paragraph	1.	Scope	1
	1.1	Purpose	1
	2.	Referenced Documents	1
	3.	Definitions (Not applicable)	2
	4.	General Requirements	2
	4.1	Legend for strip, dial or drum	2
	4.2	Engraved dials for fire control instruments	3
	4.3.1	Illustrations	3
		Dwg 264831	
		Dwg 264832	
		Dwg 264833	
		Dwg 275109	
	5.	Detail Requirements	3
	5.1	Externally illuminated sight scales and dials	3
	5.2	Sight angle scales	4
	5.3	Deflection scales	5
	5.4	Range scales	6
	5.5	Treatment of externally illuminated dials, drums, or strips after engraving	6
	5.6	Preparation of wax for filling engravings	7
	5.7	Method of filling engravings with wax	7
	5.8	When a dull background is specified	7
	5.9	Internally illuminated sight scales and dials	8
	5.10	Treatment of internally illuminated dials or drums after engraving	8
	5.11	Other acceptable methods for graduating internally illuminated dials and drums	9
	5.12	Filling the engraving of plastic and metal dials and plates	9

FIGURES

Figure	1.	Sight legend	3
	2.	Style of numerals, externally illuminated sight scales and dials	4
	3.	General forms of graduations	4
	4.	Style when the graduations take in more than the complete circumference of the drum	5
	5.	Style when the graduations take in less than the complete circumference of the drum	5
	6.	General form of graduations for range scales	6
	7.	Style of numerals, internally illuminated sight scales and dials	8

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CONTENTS (Cont)

TABLES

Page

Table I. Spacing of numerals 4

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1. SCOPE

1.1 Purpose. The purpose of this standard is to establish a uniform method of marking scales for externally and internally illuminated sights and fire control instruments, including the treatment of the surface and the filling of markings after engraving.

1.1.1 Classification.

Class 1. Externally illuminated gunsights.
Class 2. Internally illuminated gunsights.

2. APPLICABLE DOCUMENTS

2.1 The issues of the following documents, in effect on the date of invitations for bids form a part of this standard to the extent specified herein.

SPECIFICATIONS

Federal

C-B-191	Beeswax; Technical Grade
P-S-661	Solvent, Dry-Cleaning
TT-B-846	Alcohol, Butyl, Normal (For Use in Organic Coatings)
TT-E-489	Enamel, Gloss, Synthetic (For Exterior and Interior Surfaces)
TT-E-527	Enamel, Synthetic, Lustreless
TT-L-31	Lacquer, Cellulose Nitrate, Gloss
TT-L-70	Lampblack; Dry (Paint-Pigment)
TT-P-400	Pigment, Lithopone, Dry (For Protective Coating)
TT-S-271	Shellac, "Dry" (Orange) and Other Lacs
TT-T-266	Thinner; Dope and Lacquer (Cellulose-Nitrate)
VV-B-231	Benzene (Benzol), Technical

Military

JAN-A-489	Acetone (For Ordnance Use)
MIL-E-1115	Enamel, White (Formula 30) (For Naval Shipboard Use)
MIL-P-2856	Paint, Striping, Black (Formula No. 38)
MIL-C-5541	Chemical Films for Aluminum and Aluminum Alloys
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys

STANDARDS

Federal

FED. STD. No. 595 Colors

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BUREAU OF ORDNANCE DRAWINGS

264831	Dials, Typical Arrangements and Graduations
264832	Standard Dials, Details
264833	Hubs and Index Details
275109	Range, Elevation and Deflection Spot Dials

(When requesting any of the applicable documents, refer to both title and number. All requests should be made via the cognizant Government inspector. Copies of this specification and other unclassified specifications and drawings required by contractors in connection with specific procurement functions should be obtained upon application to the Commanding Officer, Naval Supply Depot (Code CDS), 5801 Tabor Avenue, Philadelphia 20, Pennsylvania. All other documents should be obtained from the procuring activity or as directed by the contracting officer.)

3. DEFINITIONS - Not Applicable

4. GENERAL REQUIREMENTS

4.1 Legend for strip, dial or drum

4.1.1 On the face of each graduation strip, dial or drum of either class of sight must be engraved a legend containing the following information arranged in accordance with Figure 1. The straight Gothic type of letter shall be used and numerals shall be as shown in Figure 2.

4.1.1.1 The caliber of the gun with which the sight is used.

4.1.1.2 The mark and modification of the sight.

4.1.1.3 The initial velocity of the projectile in feet per second.

4.1.1.4 The number and date of the Navweps Pamphlet containing the range table.

4.1.1.5 The detail drawing number, piece number and revision letter from which manufactured.

4.1.1.6 The number of the sketch on which the graduations are given.

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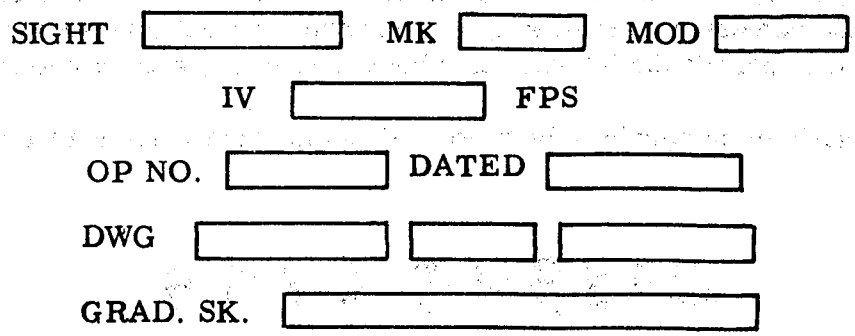


Figure 1. Sight legend

4.1.1.7 The arrangement of the information contained in Figure 1 can be varied to meet the particular application, i.e., so as to be readily read from left to right when engraved on either a drum, dial, or strip. The letters and numerals are to be engraved .125 in. high with their outlines .01 in. wide and .01 in. deep. The minimum space between the lines to be .05 in. The bottom of the recess of engraved lines can be anything from flat to a V bottom 90 degree included angle unless the ordnance drawing specifies otherwise.

4.2 Engraved dials of fire control instruments are usually internally illuminated and the following requirements for internally illuminated sights apply.

4.2.1 Bureau of Ordnance Drawings 264831, 264832, 264833, and 275109 shown on pages 13, 14, 15, and 16 illustrate the type of dials used in fire control instruments.

5. DETAILED REQUIREMENTS

5.1 Externally illuminated sight scales and dials.

5.1.2 Graduating. All graduation lines are to be a minimum of .01 in. wide and .01 in. deep. The numerals for the graduations are to be .125 in. high and their outlines to be a minimum of .01 in. wide and .01 in. deep. No graduation lines are to be closer than .05 in., except deflection scales for which the minimum space is .08 in. No numerals are to be closer than .05 in.

5.1.2.1 The graduation lines and numerals are to be engraved. The bottom of the recess of engraved lines may be anything from flat to a V bottom 90 degree included angle unless the ordnance drawing specified otherwise.

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5.1.3 When a window is used for viewing, a sight scale the size of the window or the spacing of the numerals must be such as to permit at least three numerals to be visible at all times, to permit the interpolation between numerals to be readily made.

5.1.4 The style of numerals to be used shall be in accordance with Figure 2.

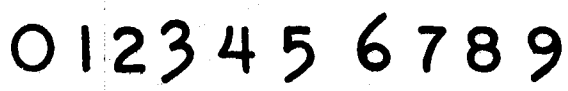


Figure 2. Style of numerals, externally illuminated sight scales and dials

5.1.5 The spacing of the numerals shall be either regular, condensed, or extra condensed as selected from Table I.

Table I. Spacing of numerals

Nom. Height	Width of Cut	Max. No. of Figures Per Inch		
		Reg.	Cond.	xCond.
1/8	.020	8	10	13
3/16	.020	5	7	8

5.2 Sight angle scales

5.2.1 Sight angle scales are graduated in minutes of arc. The minimum value graduated is 2-1/2 minutes. When this interval is less than .05 in., the minimum value becomes 5 minutes. Numerical values are to be given in multiples of 10, the minimum increment being 10 minutes or 30 minutes as required.

5.2.2 The general form of the graduations, length of lines, etc., shall be shown in Figure 3. Zero point shall be marked 2000.

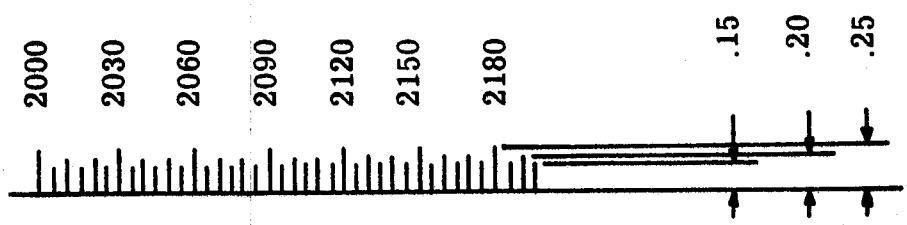


Figure 3. General forms of graduations

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5.3 Deflection scales

5.3.1 Deflection scales are graduated in mils. One mil is the angle whose tangent equals one one-thousandth (1/1000) (1 mil = 3.4377 minutes of arc). Numerals are to be provided for every fifth line to the right and left of the index line.

5.3.2 The index line for all sights is 500 mils and the form of graduations shall be as follows:

5.3.2.1 When the graduations take in more than the complete circumference of the drum they shall be in accordance with Figure 4.

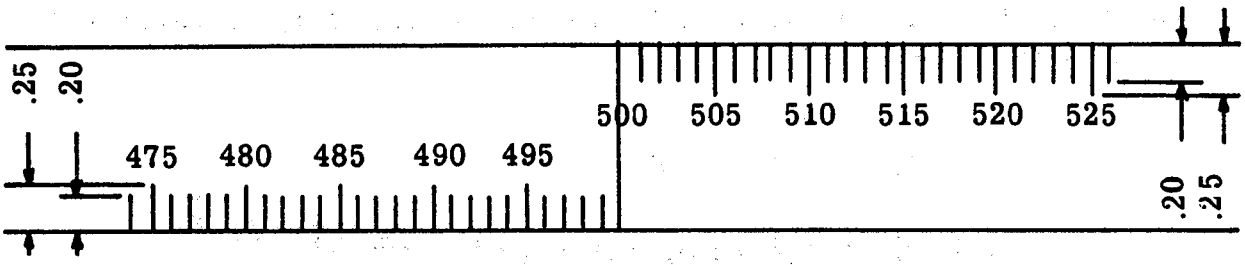


Figure 4. Style when the graduations take in more than the complete circumference of the drum

5.3.2.2 When the graduations take in less than the complete circumference of the drum they shall be in accordance with figure 5.

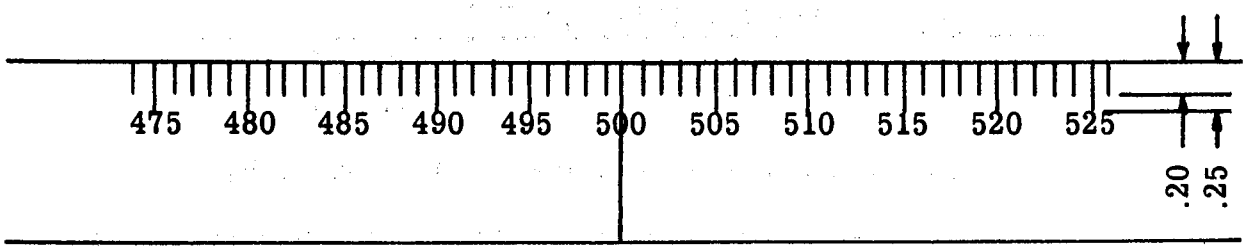


Figure 5. Style when the graduations take in less than the complete circumference of the drum

5.3.2.3 Figures 4 and 5 illustrate the method of graduating deflection drums, however; where a different design requires the use of a dial it may be necessary to rearrange the spacing and numerals consistent with the particular design.

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5.4 Range scales

5.4.1 Range scales are graduated in yards. The usual minimum value to be graduated is 100 yards. When this interval becomes less than .05 in., the minimum value becomes 200 yards. Twenty-five and fifty yard values may be shown when the provision as to viewing through a window requires them.

5.4.2 Numerical values are to be given every 100 yards, every 200 yards, every 500 yards or every 1000 yards according to the space between graduations and the size of the window, where one is used. The general form of the graduations, length of lines, etc., shall be as shown in Figure 6.

5.4.3 All range numerals are to be expressed as hundreds of yards, the last two cyphers being omitted, and near the zero point shall be engraved, "Range in hundreds of yards," in .125 in. letters.

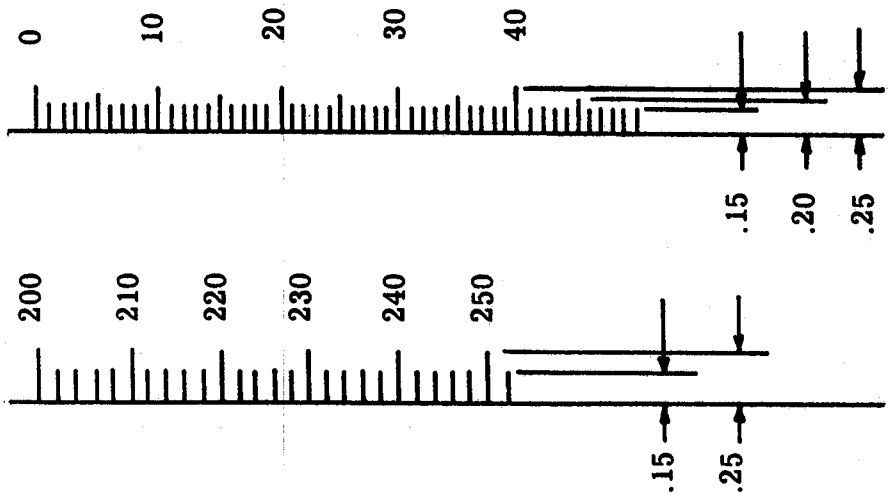


Figure 6. General form of graduations for range scales

5.4.4 On range dials, when the graduations are not on a spiral, a space of at least .50 in. must be left between the maximum range and zero range.

5.5 Treatment of externally illuminated dials, drums or strips after engraving.

5.5.1 When the engraving has been completed, the lines, letters, and numerals of dials, drums, or strips and also the descriptive marking shall be filled with white, black or red wax, or they shall be filled with white or black enamel. Metal dials, drums or strips shall have a background of dull black or dull white baking enamel sprayed on prior to filling the engraved lines, letters or numerals with enamel, Dials, drums or strips manufactured from plastic materials shall have an opaque dull colored background of black or white. The dull background may be omitted for wax filled engravings. The detail drawing of the part shall specify the treatment required.

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5.6 Preparation of the wax for filling engravings

5.6.1 The wax is prepared by melting a one pound cake (16-oz.) of Beeswax, Fed. Spec C-B-191 in a ladle and adding two ounces (2-oz.) of Lampblack Fed. Spec TT-L-70 or eight ounces (8-oz.) of either Lithopone Pigment Fed. Spec TT-P-400 or Vermilion powder commercial, depending on the color required. The melted wax must be stirred constantly while adding the color powder to obtain a smooth consistency which is poured into a mold and allowed to cool. After cooling, the sides of the mold are removed. The wax is then cut into stick form for convenience of handling usually about 1/2 in. x 1/2 in. x 4.0 in. long.

5.7 Method of filling engravings with wax

5.7.1 When filling the engravings, a stick of the wax and a hot soldering iron are held over the engravings so that the melted wax will fill the lines, letters and numerals. Another method is to heat small deflection plates or dials on a hot plate warm enough to melt the wax which is rubbed into the engravings. Excess wax is removed while the part is still warm by light scraping and after cooling by quick wiping with a cloth slightly wetted with a cleaning solvent Fed. Spec P-S-661.

5.8 When a dull background is specified

5.8.1 When a dull background is required on engraved metal surfaces, spray on one coat of Enamel, Synthetic, Lustreless, Fed. Spec TT-E-527, Color No. 37038 (Black), Fed. Std. No. 595 or one coat of Enamel Synthetic, Lustreless, Fed. Spec TT-E-527, Color No. 37875 (White), Fed. Std. No. 595. The part is then baked in an oven at a temperature of 250° F for approximately 45 minutes. After baking, remove any excess enamel from the engraving. Note: Aluminum dials, drums or strips may be anodized where required in accordance with Mil. Spec MIL-A-8625 or chemically treated in accordance with Mil. Spec MIL-C-5541 before painting. When a dull background is required on engraved plastic surfaces, no further treatment is required as plastic material is manufactured in color as specified by the procuring activity.

5.8.2 Lines, letters and numerals on dull black background are filled by brushing on White Enamel (Formula 30) Mil. Spec MIL-E-1115 or Synthetic Gloss Enamel (White) Fed. Spec TT-E-489. Lines, letters and numerals on dull white background are filled by brushing on Synthetic Gloss Enamel, Fed. Spec TT-E-489, Color No. 17038, Fed. Std. No. 595 or Black (Formula No. 38) Striping Paint, Mil. Spec. MIL-P-2856. Excess enamel is removed by first wiping the surface with a paper towel wetted with a light machine oil to prevent lifting the paint from the graduations, finally wiping with a clean paper towel. The part is then allowed to air dry.

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5.9 Internally illuminated sight scales and dials

5.9.1 Scales for internally illuminated sights shall be graduated as required by the applicable specifications for fire control systems. For metal dials or drums the style of numerals shall be in accordance with Figure 7. For plastic dials the style of numerals shall be in accordance with Figure 2.

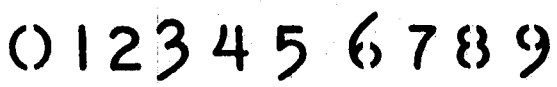


Figure 7. Style of numerals, internally illuminated sight scales and dials

5.9.2 The spacing of the numerals shall be regular, condensed or extra condensed as selected from Table I for externally illuminated sights.

5.9.3 The engraved portion of the dial or drum shall be not less than .015 in. thick. The lines and numerals shall be cut through .02 in. wide and the lines spaced not less than .07 in. on metal dials and drums. The lines and numerals on plastic dials or drums shall be at least one-half as deep as they are wide and the lines spaced not less than .05 in.

5.10 Treatment of internally illuminated dials or drums after engraving

5.10.1 Metal dials or drums shall be washed thoroughly in Benzene (Benzol) industrial grade, Fed. Spec VV-B-231 to remove dirt and finger marks and dried in an oven. Spray the face of the dial or drum with one coat of Enamel, Synthetic, Lustreless, Fed. Spec TT-E-527, Color No. 37038 (Black) Fed. Std No. 595. The part is then baked in an oven for approximately 45 minutes at a temperature of 250° F. Remove the black enamel from the engraving so as to leave all edges sharp and clean. Note: Aluminum dials or drums may be anodized where required, in accordance with Mil. Spec MIL-A-8625 or treated in accordance with Mil. Spec MIL-C-5541 before painting. Metal dials or drums may be chemically darkened. Those darkened after engraving shall require the filling of lines, letters, and numerals with translucent material. Metal dials or drums which are engraved after they have been chemically darkened may require no filling of lines, letters and numerals with translucent material.

Plastic dials or drums which require the engraved lines, letters and numerals to be filled shall be filled with the same translucent enamel used for metal dials or drums. No treatment is required prior to or after filling the engraved graduations.

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5.10.2 When either white, red or green translucent material is specified for filling lines, letters and numerals, prepare the material as follows: For white, mix equal parts by volume of Gloss Cellulose Nitrate Lacquer (White) Fed. Spec TT-L-31 Color No. 17875, Fed. Std. No. 595 with Gloss Cellulose Nitrate Lacquer (clear) Fed. Spec TT-L-31, thin this mixture with a Thinner, Fed. Spec TT-T-266, to obtain proper flowing. Fill the lines, letters or numerals with this mixture by applying to the dial or drum with a camel's hair brush. For metal dials or drums the mixture is applied to the back surface. For plastic dials or drums the mixture is applied to the engraved surfaces. Remove the excess White Lacquer from the face of the dial or drum by wiping with a cloth dampened with Acetone, Mil. Spec JAN-A-489. "CAUTION" Use Acetone sparingly on plastics. Allow to air dry and retouch to correct for air bubbles or other defects.

5.10.2.1 For red or green - Gloss Cellulose Nitrate Lacquer (Red), Fed. Spec TT-L-31 Color No. 11105, Fed. Std. No. 595 and Gloss Cellulose Nitrate Lacquer (Green), Fed. Spec TT-L-31, Color No. 14062, Fed. Std. No. 595 are prepared and applied in the same manner as the White Lacquer noted above using the same clear lacquer and thinner.

5.11 Internally illuminated dials or drums may be graduated by the photographic or silk screen methods. Care shall be exercised to insure clear lines, letters, and numerals. The style of numerals to be used shall be in accordance with figure 2.

5.12 Filling the engraving of plastic and metal dials and plates.

5.12.1 When the material used for filling in the engraving is of the same base as the background, but of different color, proceed as follows: Paint the dial or plate, prior to engraving, with the required background color selected from Fed. Std. No. 595. Allow the paint to dry thoroughly. Spray a coating of clear shellac, Fed. Spec TT-S-271, on the surface to be engraved. After engraving, spray the lines or characters using a suitable pencil gun with the desired filling color. Allow the filler color to dry, then soak the dial or plate in alcohol, Fed. Spec TT-B-846, for approximately 10 minutes. Use a soft bristle brush to flush away the excess paint. The shellac, having an alcohol base, is dissolved, leaving the desired coatings of lacquer or enamel.

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5. PROBLEM AREAS

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b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

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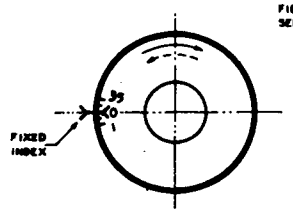
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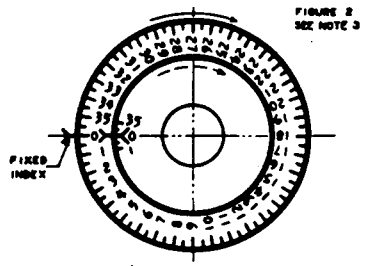
TYPICAL DIALS FOR SINGLE SPEED TRANSMISSION

FIGURE 1
SEE NOTE 3



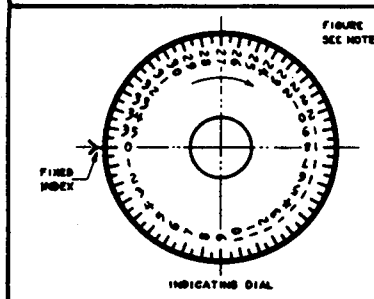
ZERO READER DIAL

FIGURE 2
SEE NOTE 3



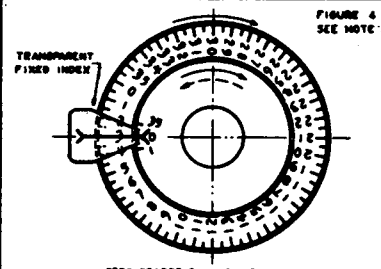
FOLLOW-THE-POINTER DIALS COMBINED WITH AN INDICATING DIAL

FIGURE 3
SEE NOTE 3



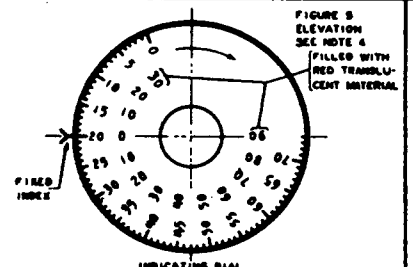
INDICATING DIAL

FIGURE 4
SEE NOTE 3



ZERO READER DIAL COMBINED WITH AN INDICATING DIAL

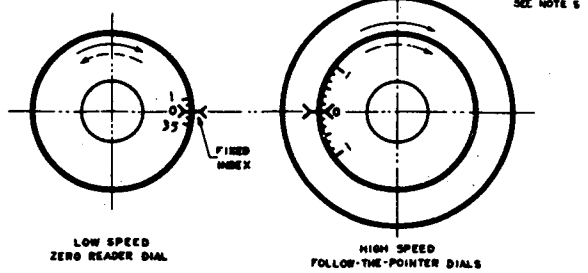
FIGURE 5
ELEVATION
SEE NOTE 4



INDICATING DIAL
INNER NUMBERS PROVIDED ONLY WHEN REQUIRED BY SPECIFICATIONS

TYPICAL DIALS FOR DOUBLE SPEED TRANSMISSION

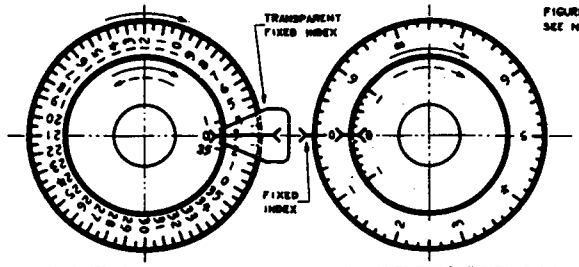
FIGURE 6
SEE NOTE 5



LOW SPEED ZERO READER DIAL

HIGH SPEED FOLLOW-THE-POINTER DIALS

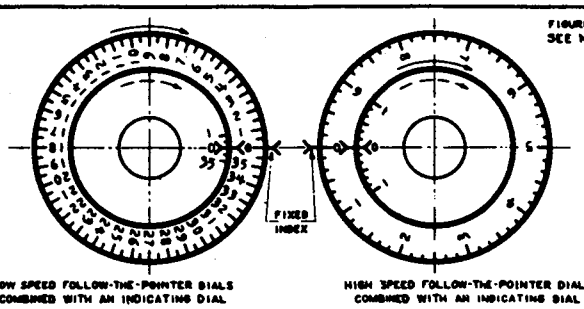
FIGURE 7
SEE NOTE 5



LOW SPEED ZERO READER DIAL COMBINED WITH AN INDICATING DIAL

HIGH SPEED FOLLOW-THE-POINTER DIALS COMBINED WITH AN INDICATING DIAL

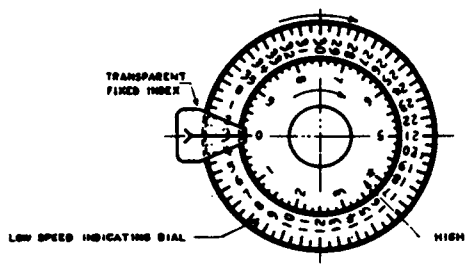
FIGURE 8
SEE NOTE 5



LOW SPEED FOLLOW-THE-POINTER DIALS COMBINED WITH AN INDICATING DIAL

HIGH SPEED FOLLOW-THE-POINTER DIALS COMBINED WITH AN INDICATING DIAL

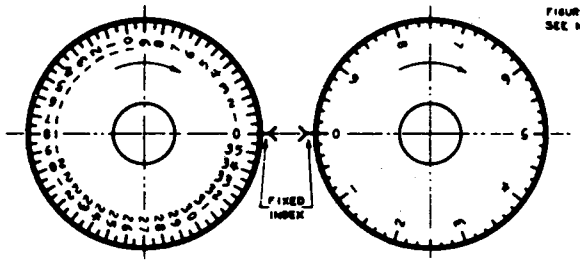
FIGURE 9
SEE NOTE 5



LOW SPEED INDICATING DIAL

HIGH SPEED INDICATING DIAL

FIGURE 10
SEE NOTE 5



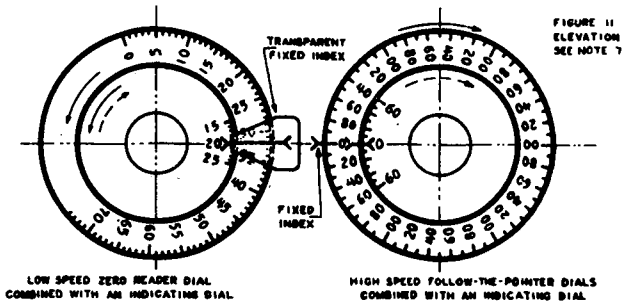
LOW SPEED INDICATING DIAL

HIGH SPEED INDICATING DIAL

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TYPICAL GRADUATIONS FOR VARIOUS ORDERS

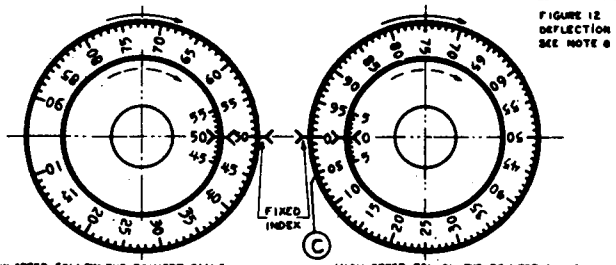
FIGURE 11
ELEVATION
SEE NOTE 7



LOW SPEED ZERO READER DIAL COMBINED WITH AN INDICATING DIAL

HIGH SPEED FOLLOW-THE-POINTER DIALS COMBINED WITH AN INDICATING DIAL

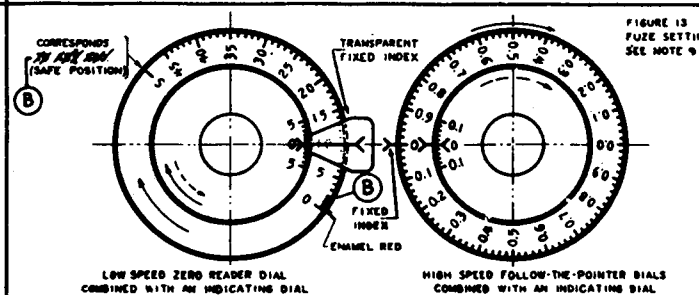
FIGURE 12
DEFLECTION
SEE NOTE 8



LOW SPEED FOLLOW-THE-POINTER DIALS COMBINED WITH AN INDICATING DIAL

HIGH SPEED FOLLOW-THE-POINTER DIALS COMBINED WITH AN INDICATING DIAL

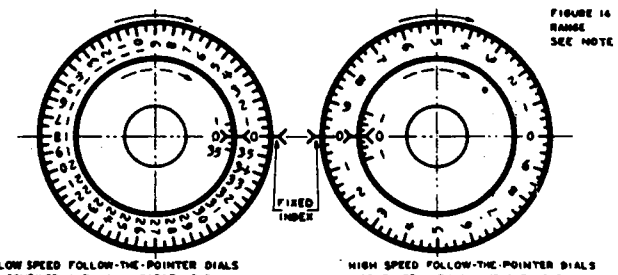
FIGURE 13
FUZE SETTING
SEE NOTE 9



LOW SPEED ZERO READER DIAL COMBINED WITH AN INDICATING DIAL

HIGH SPEED FOLLOW-THE-POINTER DIALS COMBINED WITH AN INDICATING DIAL

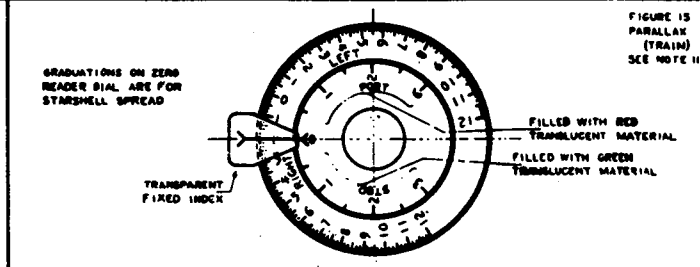
FIGURE 14
RANGE
SEE NOTE 10



LOW SPEED FOLLOW-THE-POINTER DIALS COMBINED WITH AN INDICATING DIAL

HIGH SPEED FOLLOW-THE-POINTER DIALS COMBINED WITH AN INDICATING DIAL

FIGURE 15
PARALLAX
(TRAIN)
SEE NOTE 11



LOW SPEED ZERO READER DIAL COMBINED WITH AN INDICATING DIAL

264831

NOTES:
1. AN ARROW SHOWN BY A DASH LINE INDICATES THE DIRECTION OF ROTATION FOR AN INCREASING VALUE OF TRANSMITTED OR COMPUTED QUANTITY. AN ARROW SHOWN BY A SOLID LINE INDICATES THE DIRECTION OF ROTATION FOR AN INCREASING VALUE OF RESPONSE.
2. ALL DIALS ARE TO BE CONSIDERED STANDARD INSOFAR AS THE DIRECTIONS OF ROTATION AND LOCATION OF POINTERS AND INDICES ARE CONCERNED. GRADUATIONS ARE ONLY INDICATED TO SHOW THE SEQUENCE OF GRADUATIONS FOR INCREASING QUANTITIES.
3. THE DIALS INDICATED IN FIGURES 1, 2, 3, & 4 ARE SHOWN GRADUATED FOR A "ONE" SPEED TRAIN ORDER SYSTEM.
4. THE DIAL INDICATED IN FIGURE 5 IS SHOWN GRADUATED FOR A 2 SPEED ELEVATION SYSTEM.
5. THE DIALS INDICATED IN FIGURES 6, 7, 8, 9, & 10 ARE SHOWN GRADUATED FOR A "ONE" AND 36 SPEED TRAIN ORDER SYSTEM.
6. THE COMBINATIONS OF DIALS INDICATED IN FIGURES 6, 7, 8, 9, & 10 ARE TO BE CONSIDERED STANDARD FOR DOUBLE SPEED SYSTEMS.
7. THE DIALS INDICATED IN FIGURE 11 ARE SHOWN GRADUATED FOR A 2 AND 36 SPEED ELEVATION ORDER SYSTEM.
8. THE DIALS INDICATED IN FIGURE 12 ARE SHOWN GRADUATED FOR 900 AND 100 MILS PER REV. FOR A DEFLECTION ORDER SYSTEM.
9. THE DIALS INDICATED IN FIGURE 13 ARE SHOWN GRADUATED FOR 100 AND 2 SECONDS PER REV. FOR A FUZE SETTING ORDER SYSTEM.
10. THE DIALS INDICATED IN FIGURE 14 ARE SHOWN GRADUATED FOR 36,000 AND 2,000 YARDS PER REV. FOR A RANGE ORDER SYSTEM.
11. THE DIAL INDICATED IN FIGURE 15 IS SHOWN GRADUATED FOR A 12 SPEED TRAIN PARALLAX ORDER SYSTEM.
12. TABLE OF NUMBER AND GRADUATION SPACING:

ORDER	LOW SPEED		HIGH SPEED	
	NUMBER SPACING	GRADUATION SPACING	NUMBER SPACING	GRADUATION SPACING
TRAIN	10 DEG.	5 DEG.	1 DEG.	10 MIN.
ELEVATION	500 MIN.	100 MIN.	20 MIN.	10 MIN.
DEFLECTION	50 MILS	10 MILS	5 MILS	1 MIL
FUZE SETTING	5 SECS.	1 SEC.	½ SEC.	½ SEC.
RANGE	1000 YDS.	500 YDS.	100 YDS.	25 YDS.
PARALLAX	1 DEG.	10 MIN.	—	—

13. REFERENCES.
NAVORD OSTD. NO 51 METHOD OF MARKING GRADUATED SCALES FOR SIGHTS AND FIRE CONTROL INSTRUMENTS.
BU. ORD. DR. 264832 STANDARD DIALS – DETAILS.
BU. ORD. DR. 264833 HUBS AND INDEX – DETAILS.
BU. ORD. DR. 278109 RANGE, ELEVATION AND DEFLECTION SPOT DIALS
MIL-F-18870 – SPECIFICATIONS FOR DESIGN OF FIRE CONTROL INSTRUMENTS.

(A) ERASURE REVISION:
CHANGED NUMERALS TO AGREE WITH STD. 51.

(B) ERASURE REVISION:
CHANGED ALL NUMERAL "3"S "FROM THE FLAT TOP TO THE ROUND TOP "3"S" AS PER MIL-F-18870.

REVISIONS		
LET	NUMBER	DATE
A	—	2-6-54
B	—	4-13-54
C	SEE REV NO 264831-C	7-21-54

FIRE CONTROL INSTRUMENTS

DIALS
TYPICAL ARRANGEMENTS AND GRADUATIONS

SCALE - ½

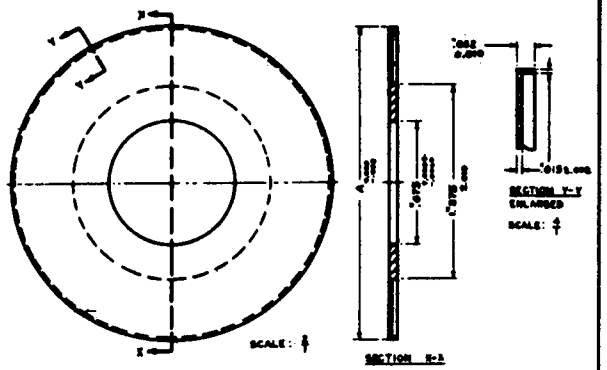
NAVY DEPARTMENT
BUREAU OF ORDNANCE
WASHINGTON, D. C
January 29, 1940

DRAWING NO.	SUPERSEDING DR. NO.	APP. SUPERSEDING DR. NO.
FIRE CONTROL EQUIPMENT		264831

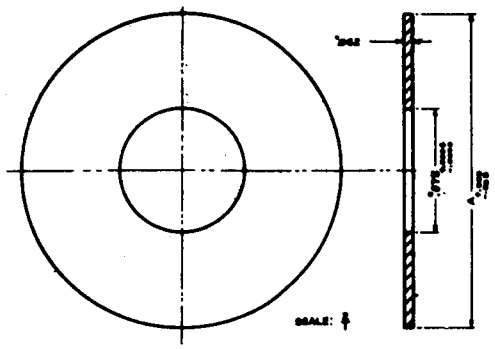
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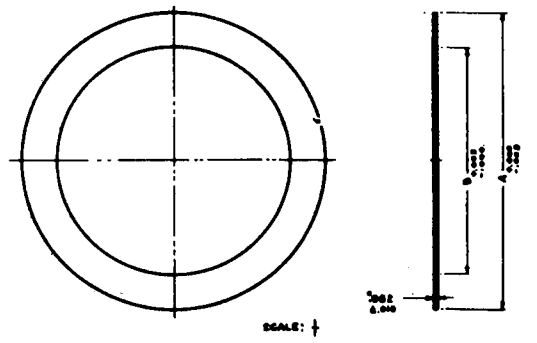
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PC. NO.	1	2	3	4	5	6	7	8	9	10
A	1.74	2.24	2.74	3.24	3.74	4.24	4.74	5.24	5.74	6.24
PC. NO.	11	12	13	14	15	16	17	18	19	20
A										



PC. NO.	21	22	23	24	25	26	27	28	29	30
A	1.74	2.24	2.74	3.24	3.74	4.24	4.74	5.24	5.74	6.24
PC. NO.	31	32	33	34	35	36	37	38	39	40
A										

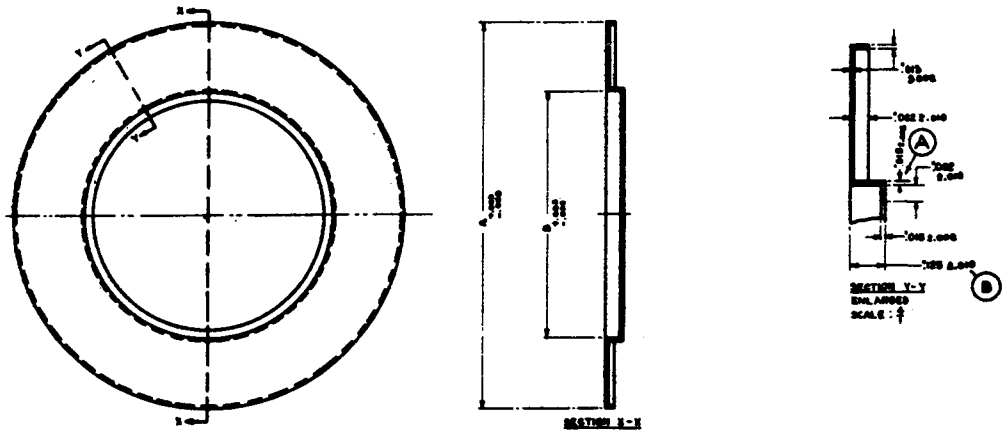


PC. NO.	41	42	43	44	45	46	47	48	49	50
A	2.74	3.24	3.74	4.24	4.74	5.24	5.74	6.24	6.74	7.24
B	1.76	2.26	2.76	3.26	3.76	4.26	4.76	5.26	5.76	6.26
PC. NO.	51	52	53	54	55	56	57	58	59	60
A										
B										

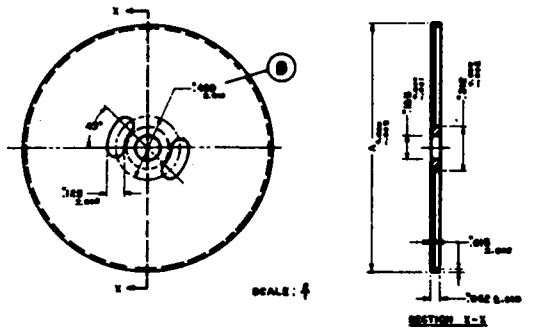
264832

NOTES:

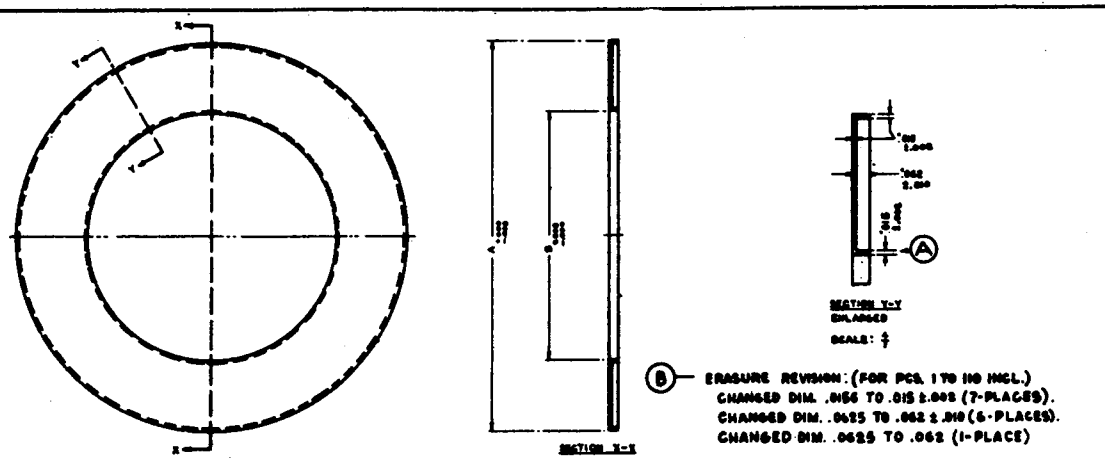
1. DIALS MUST BE FLAT, EDGE TO RUN IN TRUE PLANE .010.
2. CENTER HOLE TO BE CONCENTRIC WITH OUTSIDE DIA .003.
3. ANY GRADUATION MEASURED FROM ZERO MUST BE WITHIN 25 MINUTES OF ARC OF ITS PROPER POSITION.
4. GRADUATIONS, FIGURES, AND LETTERS FOR DIALS NOS. 1 TO 20 & 51 TO 110 INCLUSIVE, TO BE CUT THROUGH FACE OF DIAL AND FILLED WITH WHITE TRANSLUCENT PAINT OR ENAMEL.
5. FACES OF TRANSLUCENT DIALS TO BE PAINTED DULL BLACK. UNDERSIDE OF TRANSLUCENT DIALS TO BE PAINTED WITH WHITE PAINT OR ENAMEL.
6. GRADUATIONS, FIGURES & LETTERS FOR DIALS NOS. 21 TO 60 INCLUSIVE TO BE THE SAME AS FOR THE TRANSLUCENT DIALS EXCEPT THAT THEY SHALL BE .004 DEEP AND FILLED WITH WHITE PAINT OR ENAMEL.
7. OPAQUE DIALS ARE TO BE FINISHED DULL BLACK ALL OVER.
8. (B) ENGRAVING LEAVE SUFFICIENT METAL FOR A SUPPORTING WEB.
9. REFERENCES:
   (B) 264833 FOR HUBS & INDEX DETAILS.
   (C) 264831 FOR SPECIAL TOOLS FOR TYPE 1 & 5 SYNCHROS.
   (D) 264832 FOR STANDARD TESTING DIALS FOR DIAL SPINNER WRENCH FOR TYPE 1 & 5 SYNCHROS.
   (E) 272-3140 FOR 1, 2, & 3 FOR DIAL WRENCH FOR TYPE 2 SYNCHRO.
   STYLE NO. 51 MACHINE FOR MARKING GRADUATED SCALES FOR DIALS & OTHER OPTICAL INSTRUMENTS.
10. DIALS TO HAVE FINISH (C) ALL OVER.
11. FACES OF ALL DIALS TO BE FINISHED DULL BLACK.
12. GRADUATIONS, FIGURES AND LETTERS FOR DIALS NOS. 1 TO 20 AND 51 TO 110 INCLUSIVE ARE TO BE CUT THROUGH THE FACE OF THE DIAL AND FILLED WITH WHITE TRANSLUCENT PAINT OR ENAMEL, OR COLORED PAINT OR ENAMEL IF SPECIFIED. THE UNDERSIDE OF THESE DIALS IN THE AREA OF THE OPENINGS IS TO BE PAINTED WITH THE MATERIAL SPECIFIED TO FILL THE OPENINGS.
13. (B) IN ENGRAVING LEAVE SUFFICIENT METAL FOR A SUPPORTING WEB.
14. GRADUATIONS, FIGURES AND LETTERS FOR DIALS NOS. 21 TO 50 INCLUSIVE ARE TO ENGRAVED (.01 MINIMUM WIDE) AND FILLED WITH WHITE OPAQUE MATERIAL OR WITH COLORED MATERIAL IF SPECIFIED. THE UNDERSIDE OF THESE DIALS IS TO BE FINISHED DULL BLACK.
15. DIAL AND HOLE DIAMETERS GIVEN ARE TO BE OVER FINISH.
16. REFERENCES:
   (a) NAVORD OSTD 51 METHOD OF MARKING GRADUATED SCALES FOR SIGHTS AND FIRE CONTROL INSTRUMENTS.
   (b) DWG. 264833 HUBS AND INDEX - DETARS.
   (c) DWG. 264831 DIALS - TYPICAL ARRANGEMENTS AND GRADUATIONS.
   (d) DWG. 275109 RANGE, ELEVATION AND DEFLECTION SPOT DIALS.



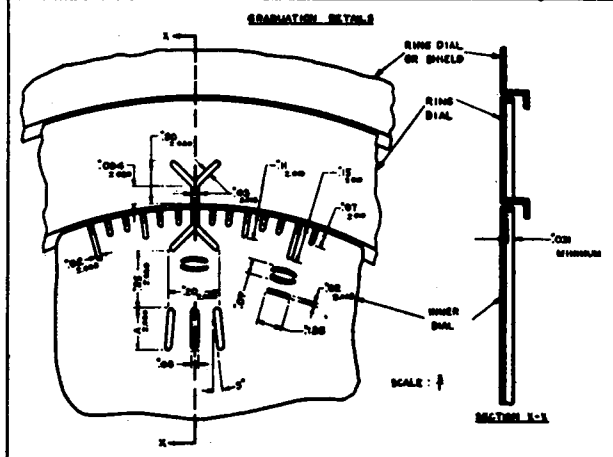
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B	1.76	2.26	2.76	3.26	3.76	4.26	4.76	5.26	5.76	6.26										



PC. NO.	101	102	103	104	105	106	107	108	109	110
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PC. NO.	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
A	2.74	3.24	3.74	4.24	4.74	5.24	5.74	6.24	6.74	7.24										
B	1.76	2.26	2.76	3.26	3.76	4.26	4.76	5.26	5.76	6.26										



PC. NO.	1, 21, & 101	2, 22, & 102	3, 23, & 103	4, 24, & 104	5 TO 20, 25 TO 40, 105 TO 110
A	.2280	.2280	.20	.30	.50

NO. TO USE	DIAL (FOR TYPE 1 SYNCHRO)		ALUMINUM ALLOY 66-A-325 Hard Temper T6
1 TO 100	DIAL (FOR TYPE 5 SYNCHRO)		ALUMINUM ALLOY 66-A-325 Hard Temper T6
PIECE NO.	NOMENCLATURE	NO. USED	MATERIAL AND SPECIFICATION NO. / WEIGHT PER PC.

"NO. REQ'D." ARE FOR

REVISIONS		
LET.	NUMBER	BASIC
A	6.2.19	30-12-40

FIRE CONTROL INSTRUMENTS

STANDARD DIALS
DETAILS
SCALE: 1 : 4 ; 3 & 4

NAVY DEPARTMENT
BUREAU OF ORDNANCE
WASHINGTON D. C.
January 29, 1940

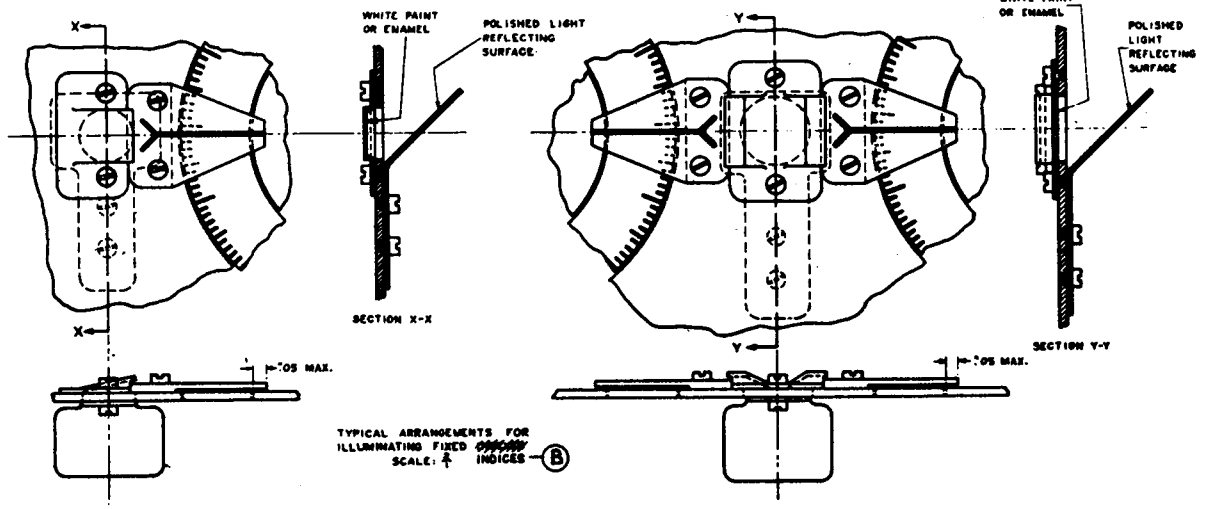
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	1/30/40		
	1/30/40		
CASTLEMAN 1-8-40	1/6/40		

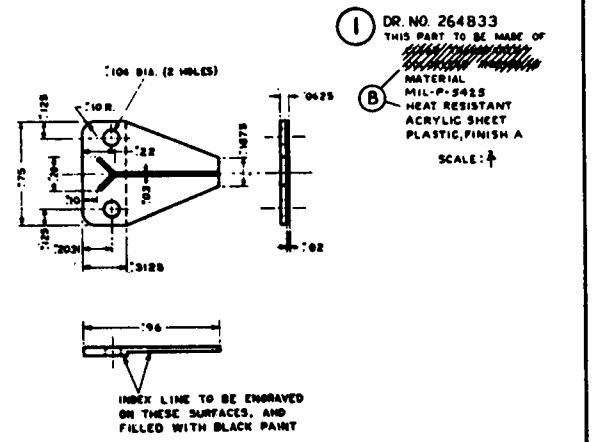
CLASSIFICATION	SUPERS. DRAWING NO.
FIRE CONTROL EQUIPMENT	264832

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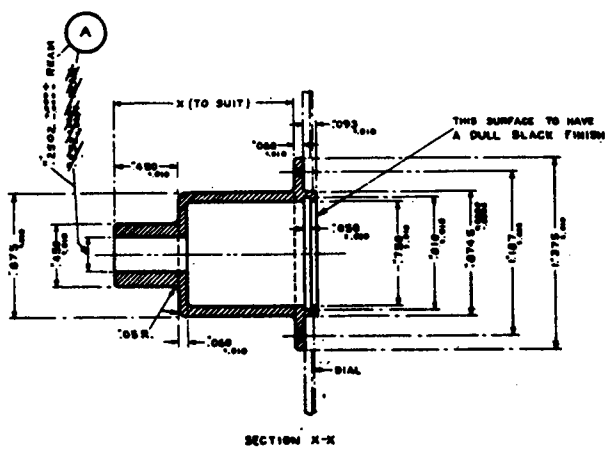
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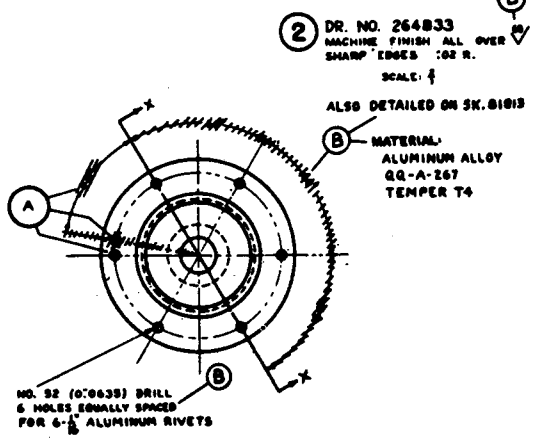


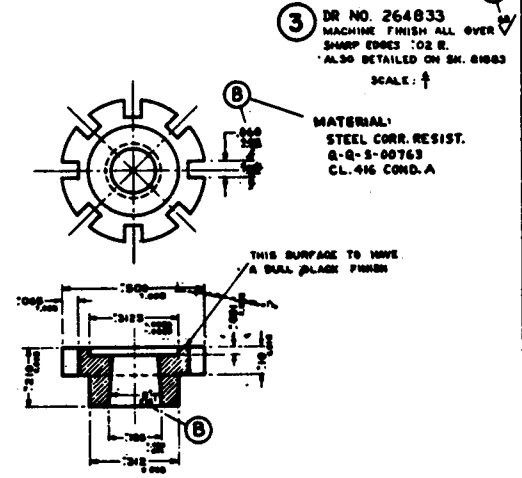


264833

NOTES:
1. SPECIFICATIONS OF LATEST ISSUE APPLY.
2. REFERENCES: ITEM 900 — B
BU. ORD. OR. 8-Z-940, SPECIAL TOOLS (DIAL WRENCH FOR TYPE 5 SYNCHRO)
BU. ORD. SK. 81883 SYNCHRO TESTING DIALS (DIAL SPANNER WRENCH FOR TYPE 1 SYNCHRO)
BU. ORD. SK. 84014 SPECIAL TOOLS FOR SYNCHRO MOTOR TYPE 1F
BU. ORD. DR. 264831 DIALS - TYPICAL ARRANGEMENTS AND GRADUATIONS
BU. ORD. DR. 264832 STANDARD DIALS - DETAILS







3	HUB (FOR TYPE 1 SYNCHRO)			
2	HUB (FOR TYPE 5 SYNCHRO)			
1	INDEX		SEE NOTE	
PIECE NO.	NOMENCLATURE	NO REQ'D	MATERIAL AND SPECIFICATION NO.	WEIGHT PER PC.

"NO. REQ'D" ARE FOR

B — SEE DETAILS

REVISIONS		
LET.	NUMBER	DATE
AA	4393	1-10-44
BM		3-24-44

FIRE CONTROL INSTRUMENTS

HUBS AND INDEX
DETAILS
SCALES: 4 & 1

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BUREAU OF ORDNANCE
WASHINGTON, D. C.
January 29, 44

CASTLEMAN	SCHULLE
7-44-4	1-44

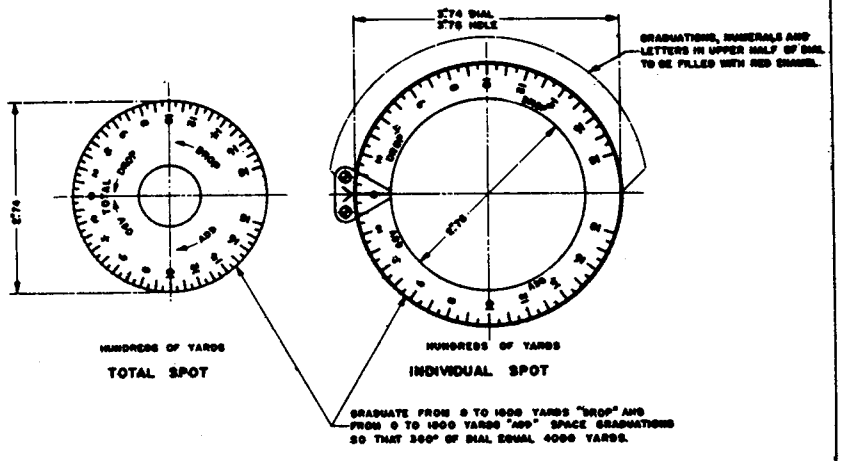
SIZE	DESCRIPTION	DWG. NO.
	FIRE CONTROL EQUIPMENT	264833

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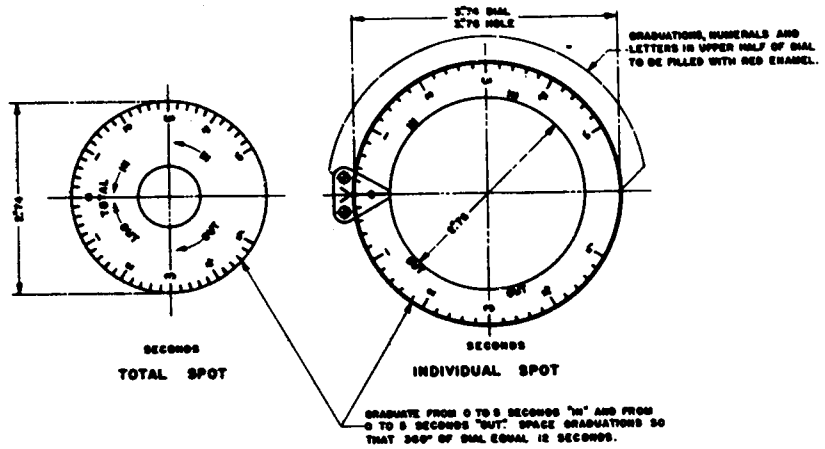
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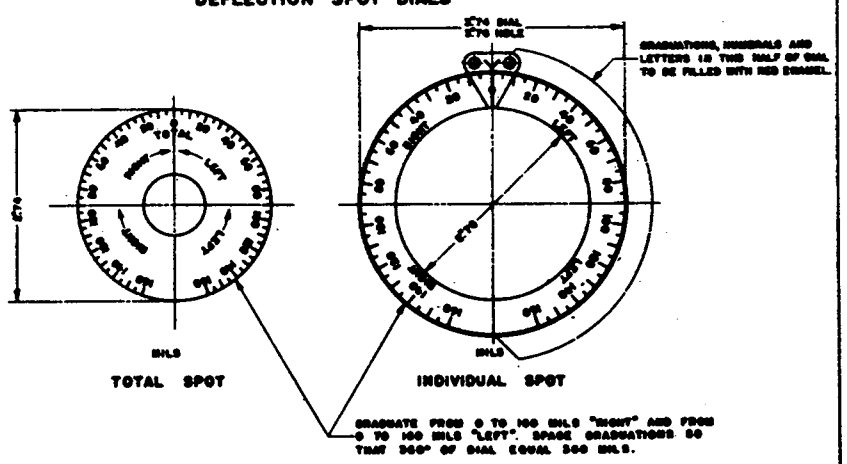
RANGE SPOT DIALS



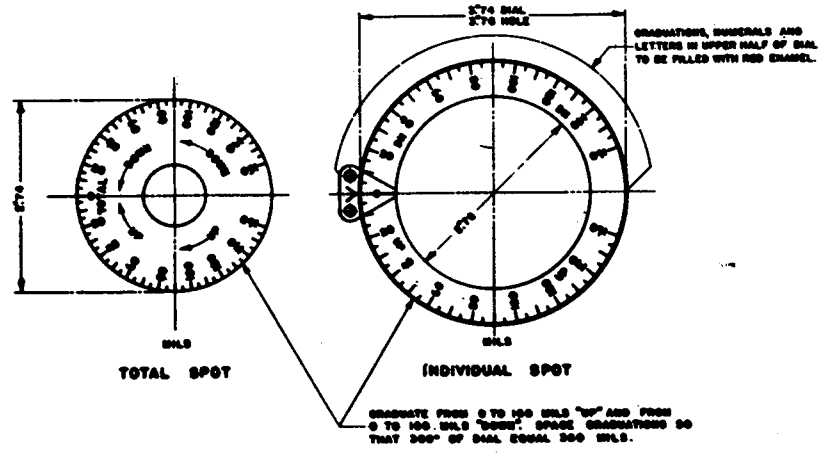
FUZE SPOT DIALS



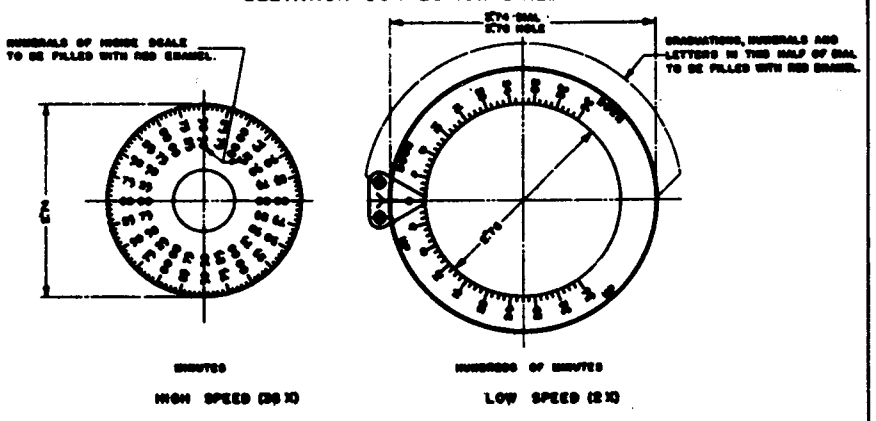
DEFLECTION SPOT DIALS



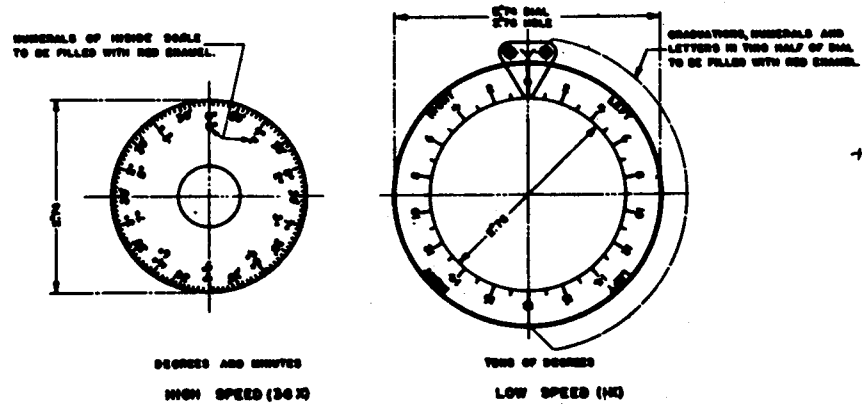
ELEVATION SPOT DIALS



ELEVATION CORRECTION DIALS



TRAIN CORRECTION DIALS



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NOTE:
(B) FOR SPECIFICATIONS GOVERNING THE MANUFACTURE OF FIRE CONTROL INSTRUMENTS SEE S.E. SYST. MIL-F-18870 SEE STD. NO.51 FOR METHOD OF MARKING GRADUATED SCALES FOR SIGHTS AND FIRE CONTROL INSTRUMENTS. FOR DETAILS OF CONSTRUCTION OF DIALS AND MARKINGS SEE DR. NOS. 264632 AND 264633.
(B) SMALL AND WHOLE NUMERALS SHOWN ARE TO BE SUCH FOR CLARITY.

(A) ERASURE REVISION:
CHANGED NUMERALS TO AGREE WITH STD. 51.
(B) ERASURE REVISION:
CHANGED ALL NUMERAL FLAT TOP "3" TO ROUND TOP'S ON RANGE SPOT DIALS & NOTE. CHANGED THE "IN" TO "DROP" & CHANGED THE "OUT" TO "ADD" (3 PLACES).

PIECE NO.	NOMENCLATURE	NO. REQ'D	MATERIAL AND SPECIFICATION NO.	WEIGHT PER PC.
	"NO. REQ'D." ARE FOR			

REVISIONS		
LET.	NUMBER	DATE
A	—	4-6-50
B	—	4-10-50

FIRE CONTROL INSTRUMENTS

RANGE, ELEVATION AND DEFLECTION SPOT DIALS.

SCALE- 1

NAVY DEPARTMENT
BUREAU OF ORDNANCE
WASHINGTON D. C.
MAY 7, 1946

DES.			
TRACED			
CHKD.			
APPD.			
O.D. 0000	4-0-00	FIRE CONTROL EQUIPMENT	275109

WARNING: THE DISCLOSURE OF NATIONAL DEFENSE SECRETS IS A PENAL OFFENSE PUNISHABLE BY A FINE, IMPRISONMENT OR BOTH.

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